

Engineering Standard Operating Procedures

Engineering Department | Nova AI Knowledge Base

1. Development Workflow

All engineering work follows the GitFlow branching strategy.

- main branch: production-ready code only. Direct pushes are prohibited.
- develop branch: integration branch for features under development.
- feature/[name] branches: created from develop for each new feature.
- hotfix/[name] branches: created from main for critical production fixes.
- All code must pass CI/CD pipeline checks before merging.

2. Code Review Policy

Every pull request (PR) must be reviewed by at least 2 engineers.

- PR description must include: purpose, approach, screenshots (if UI), and test plan.
- Reviews must be completed within 24 business hours of PR submission.
- No PR with failing tests or unresolved review comments can be merged.
- The author cannot approve their own PR.
- Use conventional commits: feat:, fix:, docs:, refactor:, test:, chore:

3. Tech Stack

The approved technology stack for all engineering projects is:

- Frontend: React 18, TypeScript, Tailwind CSS, Vite
- Backend: Node.js (Express) or Python (FastAPI)
- Database: PostgreSQL (primary), Redis (caching)
- Cloud: AWS (EC2, S3, RDS, Lambda)
- Containerization: Docker, Kubernetes
- CI/CD: GitHub Actions
- Monitoring: Datadog, Sentry

Any deviation from this stack requires CTO approval.

4. API Rate Limits

External API rate limits for partner integrations are as follows:

- Tier 1 (Standard): 1,000 requests per minute
- Tier 2 (Premium): 10,000 requests per minute
- Tier 3 (Enterprise): 100,000 requests per minute
- Burst limit: 2x the base rate for up to 30 seconds
- All limits reset on a rolling 60-second window.

- Rate limit headers must be included in all API responses: X-RateLimit-Limit, X-RateLimit-Remaining.

5. Deployment Process

Deployments follow a structured release cycle:

- Staging deployments: every Tuesday and Thursday at 2:00 PM IST
- Production deployments: every Friday at 5:00 PM IST (after QA sign-off)
- Hotfixes can be deployed any time with on-call engineer approval
- All deployments must have a rollback plan documented
- Post-deployment smoke tests must pass within 10 minutes
- Deployment freeze period: last 5 days of every quarter

6. Incident Response

For production incidents, follow the escalation matrix below:

- P0 (Critical - system down): Page on-call engineer immediately. Resolution SLA: 1 hour.
- P1 (Major - significant degradation): Notify team lead within 15 mins. SLA: 4 hours.
- P2 (Minor - partial degradation): Create Jira ticket. SLA: 24 hours.
- P3 (Low - minor issue): Add to sprint backlog. SLA: next sprint.

All P0 and P1 incidents require a post-mortem within 48 hours.

7. On-Call Rotation

Engineering maintains a 24/7 on-call rotation for production support.

- On-call shifts: Monday 9 AM to next Monday 9 AM (1 week rotations)
- Each engineer is on-call approximately once every 8 weeks
- On-call compensation: ■2,000 per week + ■500 per P0/P1 incident handled
- On-call engineers must respond to alerts within 10 minutes
- Swap requests must be submitted 48 hours in advance via the on-call tool